

Multi-Armed Bandit



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Adaptive ϵ

Limitations of ϵ -Greedy:

- Explores uniformly, not based on uncertainty
- Regret would be high
- Fixed exploration rate doesn't adapt over time

Update ϵ smartly

Adaptive ϵ

- Start with high ϵ^I – gradually reduce it to a lower value
- Any fixed lower value $\Rightarrow$ high regret
- What if $\epsilon = \frac{1}{t}$?
 - ▶ Certainly better, but not smart exploration

Introduction to UCB

Key Insight: Optimism in the Face of Uncertainty

- Use confidence bounds to guide exploration
- Explore arms that *could* be optimal
- Naturally reduce exploration as confidence increases

Upper Confidence Bound (UCB):

- Select arm with the highest upper confidence bound

UCB Algorithm

Algorithm 3: UCB1

- 1: **Input:** $\alpha > 0$
- 2: **for** $t \in \{1 \dots K\}$ **do**
- 3: Select an arm, $i_t \leftarrow t$
- 4: Observe the reward $r_{i_t, t}$
- 5: Set initial $\hat{\mu}_i \leftarrow r_{i_t, t}$
- 6: **end for**
- 7: **for** $t \in \{K + 1 \dots T\}$ **do**
- 8: Select arm $i_t \in \arg \max_{i \in [K]} \left(\hat{\mu}_{i, t} + \sqrt{\frac{\alpha \ln T}{N_i(t)}} \right)$
- 9: Observe reward $r_{i_t, t}$
- 10: Update $\hat{\mu}_{i_t}$
- 11: **end for**

UCB Regret

Theorem (Auer et al, 2002[2])

For all $K > 1$, if policy UCB1 is run on K arms having arbitrary reward distributions with support in $[0, 1]$, then its regret after T plays is at most:

$$\text{Reg}_{\text{UCB1}}(T) \leq \left[8 \sum_{i: \mu_i < \mu^*} \frac{\ln T}{\Delta_i} \right] + \left(1 + \frac{\pi^2}{3} \right) \left(\sum_{j=1}^K \Delta_j \right)$$

where $\Delta_i = \mu^ - \mu_i$ denotes the sub-optimality gap of arm i .*

Key Takeaway: The regret grows logarithmically with time T , i.e., $O(\ln T)$ (**asymptotically optimal**)



Key Steps: Decomposition

- Goal: Bound the expected number of times a suboptimal arm i is pulled, $\mathbb{E}[\mathcal{N}_i(\mathcal{T})]$.
- At any time t , if a suboptimal arm i is chosen over the optimal arm i^* , it implies:

$$\hat{\mu}_{i,t}^+ \geq \hat{\mu}_{i^*,t}^+$$

- This error occurs only if at least one of three events happens (next slide)
- We use the confidence radius:

$$c_{t,s} = \sqrt{\frac{2 \ln t}{s}}$$

Key Steps: Decomposition

Three events leading to suboptimal selection:

Event 1 **Underestimation of Optimal:**

$$\hat{\mu}_{i^*,t} \leq \mu^* - c_{t,s}$$

Event 2 **Overestimation of Suboptimal:**

$$\hat{\mu}_{i,t} \geq \mu_i + c_{t,s_i}$$

Event 3 **High Uncertainty:** The confidence interval is still wide enough to cover the gap Δ_i :

$$\mu^* < \mu_i + 2c_{t,s_i}$$



Key Steps: Bounding Probabilities

- **Bounding Events 1 & 2 (Concentration):** Using **Chernoff-Hoeffding bounds**, the probability of the empirical mean deviating by $c_{t,s}$ decays rapidly:

$$P(\text{Event 1}) \leq t^{-4} \quad \text{and} \quad P(\text{Event 2}) \leq t^{-4}$$

- Summing these over all t results in a small constant:

$$\sum_{t=1}^{\infty} 2t^{-4} \rightarrow \frac{\pi^2}{3} \quad \ominus$$

- These events contribute only $O(1)$ to the expected regret.



Key Steps: Bounding Probabilities

- **Bounding Event 3 (Deterministic Limit):** Event 3 implies that we haven't pulled arm i enough times yet.

$$\sqrt{\frac{2 \ln t}{N_{i,t}}} > \frac{\Delta_i}{2} \implies N_{i,t} < \frac{8 \ln t}{\Delta_i^2}$$

- Once arm i is sampled $I = \lceil \frac{8 \ln T}{\Delta_i^2} \rceil$ times, the confidence interval shrinks enough that this event **cannot occur**.
- **Conclusion:** The regret is dominated by the samples needed to shrink the interval, giving the $O(\ln T)$ bound:

$$\mathbb{E}[\mathcal{N}_i(T)] \leq \frac{8 \ln T}{\Delta_i^2} + \left(1 + \frac{\pi^2}{3}\right)$$

